

ENVIRONMENTAL SCIENCE AND SUSTAINABLE ENGINEERING

TECHNICAL REPORT

IMPROPER DISPOSAL OF ELECTRONIC WASTE

Solution: E-Revive – Smart E-Waste Rescue & Repurpose Platform

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Date - **8 September 2026**

TABLE OF CONTENTS

Abstract	3
1. Introduction	3
2. Problem Statement	4
3. Sustainability Analysis	5
4. Proposed Engineering/IT Solution – E-Revive	6
5. Feasibility Analysis	9
6. Expected Environmental Benefits	11
7. Justification of the Proposed Solution	11
8. Conclusion	12
References & Sources.....	13

ABSTRACT

Electronic waste (e-waste) is one of the fastest-growing waste streams in the world. Rapid replacement of smartphones, computers, chargers, appliances and other electronic products creates a growing quantity of discarded equipment. When these products are mixed with ordinary waste, stored indefinitely, burned, dismantled without safeguards or sent to unverified handlers, hazardous substances may enter air, soil and water, while valuable materials are lost. The Global E-waste Monitor reported 62 million tonnes of e-waste in 2022, with only 22.3% documented as formally collected and recycled.

This report proposes E-Revive – Smart E-Waste Rescue & Repurpose Platform, a responsive website designed around a simple principle: an old electronic device should be treated first as a resource for reuse, repair, repurposing or responsible recycling, rather than immediately as waste. The platform would allow a user to describe or select an unwanted device, receive suggested second-life applications, learn basic safe-handling steps, locate an appropriate collection/recycling option, and record the device's final pathway. The central innovation is the combination of a second-life recommendation system with responsible disposal guidance.

The solution supports circular-economy thinking by extending product life, encouraging repair and reuse, reducing avoidable disposal, and directing unavoidable e-waste towards authorised channels. It aligns particularly with SDG 12 on responsible consumption and production, while also supporting SDGs 9, 11 and 13. The report evaluates the concept through environmental management principles, a simplified life-cycle perspective, feasibility analysis, expected environmental benefits and implementation considerations.

1. INTRODUCTION

Modern society depends on electronic and electrical equipment for communication, education, healthcare, transportation, banking, entertainment and industrial activity. The rapid development of technology has made devices more capable and accessible, but it has also shortened the period for which many products are perceived as useful. A device may be discarded because of a minor defect, battery degradation, outdated software, cosmetic damage or the purchase of a newer model even when some of its components remain functional.

E-waste is not ordinary household waste. It may contain useful metals such as copper, aluminium, iron and precious metals, together with plastics, glass and other materials. At the same time, certain electronic components can contain or generate hazardous substances. WHO notes that unsound e-waste recycling can release substances including

lead, mercury, cadmium, nickel, brominated flame retardants and other pollutants into the environment.

The challenge is therefore both environmental and technological. A successful solution should not only tell people to recycle; it should influence the decision made before recycling. If a device can be repaired or reused safely, its useful life can be extended. If it cannot be reused, it should be channelled to an authorised collection or recycling route. E-Revive is proposed as a digital platform to connect these decisions in one simple user journey.

2. PROBLEM STATEMENT

The environmental problem addressed in this report is the improper disposal of electronic waste. Improper disposal occurs when unwanted electrical and electronic equipment is thrown into mixed municipal waste, dumped in open areas, burned, dismantled using unsafe methods, stored without a planned recovery route, or handed to unverified operators. The problem is intensified by limited public awareness, fragmented collection systems, uncertainty about where to dispose of specific devices and the perception that an old device has no value.

2.1 Major Causes

- Rapid technological change and frequent replacement of electronic products.
- Low awareness of the environmental and health risks associated with improper e-waste handling.
- Lack of convenient information about repair, reuse, take-back and authorised recycling options.
- Devices are often discarded because of one failed component even when other parts remain useful.
- Informal recovery practices may use unsafe dismantling, burning or chemical extraction methods.
- Consumers may not know how to remove personal data before giving away a phone, laptop or storage device.
- Collection and recycling information is often scattered across different organizations and locations.

2.2 Environmental and Health Impacts

Improperly managed e-waste can contaminate environmental media and increase exposure to hazardous substances. Open burning may release toxic fumes and contaminated particulates. Dumping can allow contaminants to enter soil and water. Unsafe dismantling can expose workers to dust, sharp components and hazardous

materials. WHO identifies children and pregnant women as particularly vulnerable to harmful exposures associated with unsound e-waste recycling.

Improper disposal also causes resource loss. Electronics contain materials that required energy, mining, processing and transportation before becoming part of a product. Recovering these materials through environmentally sound recycling can reduce the need for some virgin-resource extraction. The UN notes that e-waste is associated with the loss of scarce and valuable raw materials and that circular-economy approaches are important for sustainable consumption and production.

2.3 Engineering Problem Definition

The engineering problem can be expressed as follows: How can a low-cost, user-friendly digital system encourage people to keep electronic products in productive use for longer and, when reuse is no longer practical, guide them towards safe and traceable end-of-life management?

2.4 Existing Gap

Existing approaches often focus on collection or recycling after a product becomes waste. E-Revive addresses the earlier decision point. It asks what useful function can be recovered before the device enters the waste stream. This makes the concept different from a simple e-waste collection directory.

3. SUSTAINABILITY ANALYSIS

3.1 Relevant Sustainable Development Goals

SDG	Relevance to E-Revive	Contribution
SDG 12 – Responsible Consumption and Production	Primary SDG	Promotes prevention, reuse, repair, recycling and informed consumption. Target 12.5 specifically calls for substantially reducing waste through prevention, reduction, recycling and reuse.
SDG 9 – Industry, Innovation and Infrastructure	Digital innovation	Uses a responsive web platform and data-driven recommendations to support sustainable technology use and circular innovation.
SDG 11 – Sustainable Cities and Communities	Urban waste management	Can help households and institutions channel electronic discards away from mixed municipal waste.
SDG 13 – Climate Action	Indirect contribution	Longer product lifetimes and material recovery can reduce some upstream energy and emissions associated with new production and raw-material extraction.

3.2 ISO 14001 Perspective

ISO 14001 is the global standard for environmental management systems (EMS), helping organizations manage environmental aspects and improve performance. The 2026 edition outlines a framework for compliance and objective setting. E-Revive is designed to assist in ISO 14001-style management by offering a structured approach to identify and control the environmental impact of discarded electronics.

- Environmental aspect: generation and handling of obsolete electronic equipment.
- Operational control: guidance that prioritises reuse/repair and directs residual e-waste to appropriate handlers.
- Awareness: educational content that improves user competence and environmental responsibility.
- Monitoring: dashboard indicators such as devices listed, devices reused, devices donated, collection requests and estimated e-waste diverted.
- Continual improvement: use of user feedback and outcome data to refine recommendations.

3.3 Simplified Life-Cycle Assessment (LCA) Perspective

A formal ISO-compliant LCA would require measured inventory data for materials, energy, transport, use and end-of-life stages. Such data are outside the scope of this student-level conceptual project. However, a simplified life-cycle perspective can still identify where E-Revive creates value:

1. Manufacturing stage – extending the useful life of an existing device can delay demand for a replacement device.
2. Use stage – repair and repurposing may keep components productive without complete replacement.
3. Collection stage – the platform can reduce uncertainty about where the item should go.
4. End-of-life stage – non-reusable devices can be directed towards environmentally sound recycling.
5. Feedback stage – outcome data can help improve future collection and reuse decisions.

4. PROPOSED ENGINEERING/IT SOLUTION: E-REVIVE

4.1 Solution Overview

E-Revive – Smart E-Waste Rescue & Repurpose Platform is a responsive website that helps users discover useful second-life applications for old electronic devices and their components instead of throwing them away. The key idea is to intervene before an item becomes unmanaged waste.

The platform would act as a decision-support layer between the user and the e-waste management ecosystem. Instead of presenting recycling as the only option, it would use a simple device profile to suggest a hierarchy of actions: **REUSE** → **REPAIR** → **REPURPOSE** → **DONATE/SELL** → **RESPONSIBLE RECYCLE**. The final option depends on the device condition, safety, data-storage characteristics and local availability of appropriate collection channels.

4.2 Core Idea: From 'Waste' to 'Resource'

A broken laptop may still contain a working display, RAM, storage drive, keyboard, fan or power adapter. An old smartphone may be usable as a security camera, music player, digital clock or learning device after appropriate software and privacy checks. A retired router may be repurposed for network practice or a lab environment. A computer cabinet, cooling fan or power supply may also have educational or engineering uses when handled safely.

E-Revive does not encourage unsafe self-repair or dismantling. Instead, it separates low-risk reuse ideas from tasks requiring a trained technician. The website would clearly mark safety-sensitive activities and provide a responsible recycling pathway when reuse is unsuitable.

4.3 Proposed User Workflow

User opens site	Selects old device	Enters condition	E-Revive suggests	User chooses pathway	Collection / reuse action	Outcome recorded
↓	↓	↓	↓	↓	↓	✓

4.4 Main Website Modules

Module	Purpose
Device Selector	User selects categories such as phone, laptop, desktop, monitor, printer, charger, router, small appliance or component.
Condition Checker	Simple questions classify the item as working, partly working, repairable, damaged or unsafe to reuse.
Second-Life Recommendation Engine	Maps the device and condition to possible reuse, repair or repurpose ideas.
Safe Handling & Data Privacy	Provides reminders for battery safety, personal-data backup/erasure and precautions against unsafe dismantling.
Responsible Disposal Finder	Displays appropriate collection/recycling guidance and, in a real deployment, could connect to verified local recyclers or take-back points.
Impact Dashboard	Shows the user's contribution through devices reused, donated, repaired or responsibly channelled.
Awareness Hub	Short articles, quizzes and infographics explain e-waste, circular economy and sustainable consumption.

4.5 Technical Concept and Data Flow

For a future implementation, the website can be built using standard responsive web technologies. The front end may use HTML, CSS and JavaScript or a modern framework such as React. A lightweight backend can store device categories, recommendation rules, collection-point information and anonymised impact statistics. A relational or document database can be used depending on the final implementation.

The first version does not require artificial intelligence. A rule-based recommendation engine is sufficient for a second-year engineering prototype. For example, if a laptop powers on but has a weak battery, the system can recommend battery replacement, use as a desktop-connected device, donation after repair, or responsible recycling if the battery is damaged. A later version could use machine learning or image classification to identify devices from photographs.

4.6 Recommendation Logic

1. Identify the device category.
2. Ask whether it powers on and whether major functions work.
3. Check whether repair is potentially practical.
4. Check for safety conditions such as swelling batteries, exposed wires, broken mains components or water damage.
5. Suggest safe second-life applications where appropriate.
6. If reuse/repair is unsuitable, suggest donation/sale or responsible recycling.
7. Record the selected outcome for impact reporting.

4.7 Example Use Case

Consider a student who has an old laptop that is slow but still operational. The student selects 'Laptop' and answers that the display, keyboard and charging system work. E-Revive could recommend: (1) clean the system and replace a storage drive if appropriate; (2) use it for programming practice or as a lightweight study computer; (3) donate it to a student or community organisation after data removal; or (4) if the machine is beyond economical repair, use a verified e-waste recycling route. The platform therefore prevents the immediate jump from 'old' to 'waste'.

4.8 Safety and Privacy by Design

- Never instruct users to open mains-powered equipment unless they have appropriate technical competence.
- Flag swollen, leaking or damaged batteries for professional handling rather than DIY repair.
- Encourage backup and secure erasure of personal data before donation, resale or recycling.

- Avoid collecting unnecessary personal information.
- Use verified or authoritative sources for recycler/collection information in a real deployment.
- Clearly distinguish educational repurposing ideas from activities requiring trained technicians.

4.9 Uniqueness of the Proposed Solution

The distinctive feature of E-Revive is the 'repurpose-first' decision layer. A conventional directory may answer 'Where can I recycle this?' E-Revive asks a prior question: 'Can this device still provide useful value?' This supports the waste hierarchy and circular-economy principle of retaining products and materials at their highest practical value before recycling.

5. FEASIBILITY ANALYSIS

5.1 Technical Feasibility

The concept is technically feasible using widely available web-development technologies. A prototype can be developed without specialised hardware. The recommendation engine can initially use a database of devices and rule-based decision trees. Responsive design allows the website to work on mobile phones, tablets and desktop computers. APIs or map services may later be integrated to display verified collection locations.

5.2 Economic Feasibility

A student prototype can be developed at very low cost using open-source software, a local development environment and free or low-cost hosting. The major cost in a production system would be maintaining accurate collection-point data, hosting, security, domain services and partnerships with authorised recyclers. Because the platform is informational and coordination-oriented, its initial capital requirement is much lower than that of a physical recycling facility.

5.3 Operational Feasibility

The proposed user journey is intentionally simple: select a device, answer a few questions, view options and choose an appropriate pathway. This makes it suitable for households, schools, colleges, offices and small organisations. The platform could also be used in campus e-waste drives, where students register devices and receive guidance before handing them over.

5.4 Legal and Regulatory Feasibility in India

India's E-Waste (Management) Rules, 2022 came into force on 1 April 2023. The rules establish responsibilities for manufacturers, producers, refurbishers, dismantlers and recyclers, including Extended Producer Responsibility (EPR). CPCB explains that the rules encourage reuse through refurbishing certificates and require EPR obligations to be met through registered recyclers. Therefore, a real E-Revive deployment should not claim that every listed recycler is authorised without verification; its database should be connected to current official or verified information.

5.5 Social Feasibility

The platform has potential social value because it turns an environmental instruction into a practical action. Students and households are more likely to participate when they receive a clear answer about what to do with a specific device. The awareness component can also encourage responsible consumption and reduce the idea that upgrading automatically means discarding.

5.6 Risk Analysis and Mitigation

Risk	Potential Effect	Mitigation
Incorrect reuse advice	Safety or device damage	Use conservative rules; flag unsafe cases and recommend professionals.
Outdated collection data	User sent to unsuitable location	Periodic verification and official-source updates.
Data privacy	Personal information exposure	Collect minimum data and provide clear deletion/privacy controls.
Low user participation	Limited environmental impact	Gamification, campus campaigns and simple user flows.
Greenwashing / false impact claims	Loss of credibility	Label estimates clearly and report only measurable outcomes.

6. EXPECTED ENVIRONMENTAL BENEFITS

The benefits of E-Revive arise mainly from changing the fate of electronic products. The website itself does not recycle materials; its environmental function is to improve decisions, extend product use and guide residual waste to appropriate channels.

Expected Benefit	How E-Revive Contributes
Waste prevention	Some devices may be reused or repaired instead of being discarded immediately.
Product-life extension	Second-life applications can postpone replacement and disposal.
Resource conservation	Reuse and recycling can retain valuable materials within productive or recovery cycles.
Reduced unsafe disposal	Clear guidance can discourage dumping, open burning and unverified handling.
Improved awareness	Users learn why e-waste is different from ordinary waste.
Better traceability	A future impact dashboard can record outcomes and identify where interventions are succeeding.
Support for circular economy	The platform encourages keeping products and components at higher-value uses for longer.

6.1 Suggested Performance Indicators

- Number of users completing the e-waste decision workflow.
- Number of devices recommended for reuse, repair, donation and recycling.
- Number of devices actually reported as reused or responsibly channelled.
- Estimated mass of e-waste diverted from mixed waste, where reliable weight data are available.
- Number of verified collection/recycling partners.
- User awareness improvement measured through optional pre- and post-use quizzes.

7. JUSTIFICATION OF THE PROPOSED SOLUTION

E-Revive is justified because improper e-waste disposal is partly a decision and information problem. Regulations and recycling infrastructure are necessary, but a person still needs to know what to do when a device becomes unwanted. A platform that

combines second-life ideas, safety guidance and responsible disposal information can close this practical gap.

The proposal also follows the hierarchy of preferred waste management actions. Preventing waste is generally preferable to managing waste after it is created. Reuse and repair preserve more of the original product than material recycling alone. Recycling remains essential for devices that cannot safely or economically be reused, but it should be a controlled end-of-life pathway.

The solution demonstrates the application of software engineering to a real environmental problem. It can be implemented as a manageable prototype while still having a clear pathway to a larger system. Students can contribute through front-end design, database development, rule-engine development, environmental content preparation, user testing and impact measurement.

Most importantly, E-Revive avoids presenting technology as the complete solution to e-waste. The website is a coordination and behaviour-change tool. Its environmental effectiveness depends on reliable information, user participation, safe handling practices and links to legitimate recycling/refurbishment systems.

8. CONCLUSION

Improper disposal of electronic waste is a growing environmental challenge because electronic products combine valuable resources with components that can create health and environmental risks when managed unsafely. Global e-waste generation is increasing rapidly, while formal recycling is not keeping pace. The problem therefore requires both stronger waste-management systems and better decisions by consumers and institutions.

This report proposed E-Revive – Smart E-Waste Rescue & Repurpose Platform as a unique IT-based response. Its main idea is to intervene before an unwanted electronic device becomes unmanaged waste. By asking users to identify the device and its condition, the platform can recommend safe second-life options such as repair, reuse, donation or repurposing, while providing a responsible recycling pathway when necessary.

The concept supports SDG 12 and contributes to broader sustainable-development goals. It is compatible with environmental management principles reflected in ISO 14001 and can be evaluated through a life-cycle perspective. In India, the platform should align with

the E-Waste (Management) Rules, 2022 and use verified information regarding registered recyclers and EPR processes.

The proposed solution is technically, economically and operationally feasible as a student-level web project. Its greatest potential lies in combining environmental awareness with a practical action pathway. If implemented responsibly and connected to verified collection and recycling networks, E-Revive can help shift the mindset from 'old electronics are waste' to 'old electronics are resources that should be reused, recovered or safely recycled.

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